

Determine the amount of excavation needed for the continuous footings in Figures 9.54 and 9.55. A one-foot working distance is required and the excavation will be sloped 1:1

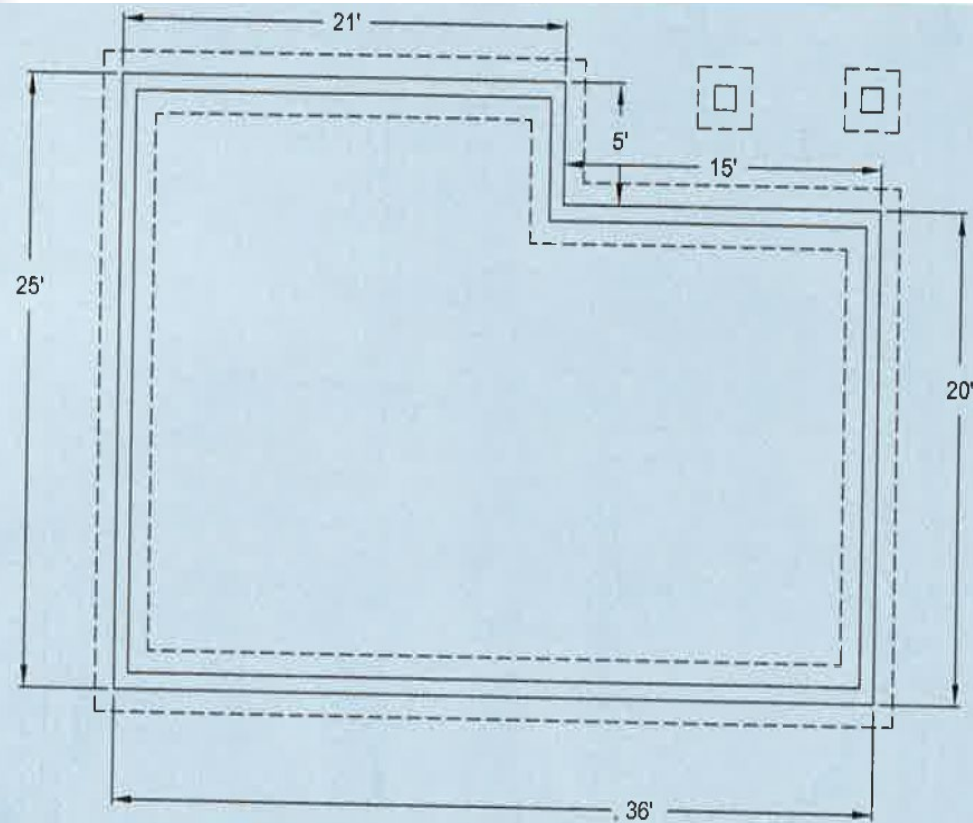


FIGURE 9.54. Foundation Plan View.

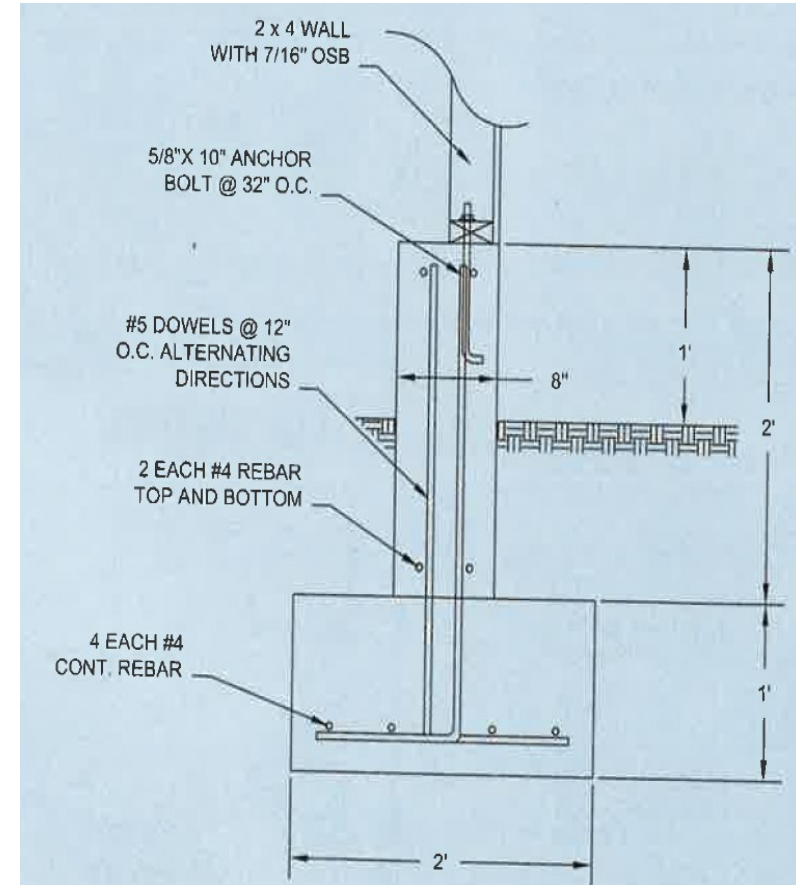


FIGURE 9.55. Wall Section.

Determine the amount of excavation needed for the continuous footings in Figures 9.54 and 9.55. A one-foot working distance is required and the excavation will be sloped 1:1

$$\text{Depth} = 2' - 1' + 1' = 2'$$

$$\text{Perimeter} = 25' + 21' + 5' + 15' + 20' + 36' = 122'$$

$$\text{Distance}_{\text{Average}} = 1' + 2' + 1' + 2' = 6'$$

General excavation =

$$122' \times 6' \times 2' = 1,464 \text{ bcf} / 27 \text{ cf per cy} = 54.2 \text{ bcy}$$

Determine the amount of backfill needed for the continuous footing and foundation in the previous problem.

Volume of Footing and Foundation =

$$(2' \times 1' + 1' \times 8'') \times 122' = 325.34 \text{ sf} / 27 \text{ cf per cy} = 12.0 \text{ cy}$$

The backfill will be in compacted cubic yards

$$\text{Backfill (ccy)} = 54.2 \text{ bcy} - 12.0 \text{ cy} = 42.2 \text{ ccy}$$

Spread Footings

Volume

=Average length of cut
x Average width of cut
x depth of cut